

Number Theory

Chapter 4

With Question/Answer Animations

Section Summary

- Division
- Division Algorithm
- Modular Arithmetic

DIVISION

Definition: If a and b are integers with $a \neq 0$, then a divides b if there exists an integer c such that $b = ac$.

- When a divides b we say that a is a *factor* or *divisor* of b and that b is a multiple of a .
- The notation $a \mid b$ denotes that a divides b .
- If $a \mid b$, then b/a is an integer.
- If a does not divide b , we write $a \nmid b$.

Example: Determine whether $3 \mid 7$ and whether $3 \mid 12$.

EXAMPLE 3 What are the quotient and remainder when 101 is divided by 11?

Solution: We have

$$101 = 11 \cdot 9 + 2.$$

Hence, the quotient when 101 is divided by 11 is $9 = 101 \text{ div } 11$, and the remainder is $2 = 101 \text{ mod } 11$. ◀

EXAMPLE 4 What are the quotient and remainder when -11 is divided by 3?

Solution: We have

$$-11 = 3(-4) + 1.$$



Hence, the quotient when -11 is divided by 3 is $-4 = -11 \text{ div } 3$, and the remainder is $1 = -11 \text{ mod } 3$.

Note that the remainder cannot be negative. Consequently, the remainder is *not* -2 , even though

$$-11 = 3(-3) - 2,$$

because $r = -2$ does not satisfy $0 \leq r < 3$. ◀

PROPERTIES OF DIVISIBILITY

Theorem 1: Let a , b , and c be integers, where $a \neq 0$.

- i. If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;
- ii. If $a \mid b$, then $a \mid bc$ for all integers c ;
- iii. If $a \mid b$ and $b \mid c$, then $a \mid c$.

Proof: (i) Suppose $a \mid b$ and $a \mid c$, then it follows that there are integers s and t with $b = as$ and $c = at$. Hence,

$$b + c = as + at = a(s + t). \quad \text{Hence, } a \mid (b + c)$$



Corollary: If a , b , and c be integers, where $a \neq 0$, such that $a \mid b$ and $a \mid c$, then $a \mid mb + nc$ whenever m and n are integers.

DIVISION ALGORITHM

- When an integer is divided by a positive integer, there is a quotient and a remainder. This is traditionally called the “Division Algorithm,” but is really a theorem.

Division Algorithm: If a is an integer and d a positive integer, then there are unique integers q and r , with $0 \leq r < d$, such that $a = dq + r$ (proved in Section 5.2).

- d is called the *divisor*.
- a is called the *dividend*.
- q is called the *quotient*.
- r is called the *remainder*.

Definitions of Functions
div and **mod**

$$q = a \text{ div } d$$
$$r = a \text{ mod } d$$

Examples:

- What are the quotient and remainder when 101 is divided by 11?

Solution: The quotient when 101 is divided by 11 is $9 = 101 \text{ div } 11$, and the remainder is $2 = 101 \text{ mod } 11$.

- What are the quotient and remainder when -11 is divided by 3?

Solution: The quotient when -11 is divided by 3 is $-4 = -11 \text{ div } 3$, and the remainder is $1 = -11 \text{ mod } 3$.

CONGRUENCE RELATION

Definition: If a and b are integers and m is a positive integer, then a is *congruent to b modulo m* if m divides $a - b$.

- The notation $a \equiv b \pmod{m}$ says that a is congruent to b modulo m .
- We say that $a \equiv b \pmod{m}$ is a *congruence* and that m is its *modulus*.
- Two integers are congruent mod m if and only if they have the same remainder when divided by m .
- If a is not congruent to b modulo m , we write

$$a \not\equiv b \pmod{m}$$

Example: Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6.

Solution:

- $17 \equiv 5 \pmod{6}$ because 6 divides $17 - 5 = 12$.
- $24 \not\equiv 14 \pmod{6}$ since $24 - 14 = 10$ is not divisible by 6.

MORE ON CONGRUENCES

Theorem 4: Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Proof:

- If $a \equiv b \pmod{m}$, then (by the definition of congruence) $m \mid a - b$. Hence, there is an integer k such that $a - b = km$ and equivalently $a = b + km$.
- Conversely, if there is an integer k such that $a = b + km$, then $km = a - b$. Hence, $m \mid a - b$ and $a \equiv b \pmod{m}$. ◀

The Relationship between $(\text{mod } m)$ and $\text{mod } m$ Notations

- The use of “mod” in $a \equiv b \pmod{m}$ and $a \text{ mod } m = b$ are different.
 - $a \equiv b \pmod{m}$ is a relation on the set of integers.
 - In $a \text{ mod } m = b$, the notation **mod** denotes a function.
- The relationship between these notations is made clear in this theorem.
- **Theorem 3:** Let a and b be integers, and let m be a positive integer. Then $a \equiv b \pmod{m}$ if and only if $a \text{ mod } m = b \text{ mod } m$. (*Proof in the exercises*)

Congruences of Sums and Products

Theorem 5: Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$$a + c \equiv b + d \pmod{m} \text{ and } ac \equiv bd \pmod{m}$$

Proof:

- Because $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, by Theorem 4 there are integers s and t with $b = a + sm$ and $d = c + tm$.
- Therefore,
 - $b + d = (a + sm) + (c + tm) = (a + c) + m(s + t)$ and
 - $bd = (a + sm)(c + tm) = ac + m(at + cs + stm)$.
- Hence, $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Example: Because $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$, it follows from Theorem 5 that

$$18 = 7 + 11 \equiv 2 + 1 = 3 \pmod{5}$$

$$77 = 7 \cdot 11 \equiv 2 \cdot 1 = 2 \pmod{5}$$

ALGEBRAIC MANIPULATION OF CONGRUENCES

- Multiplying both sides of a valid congruence by an integer preserves validity.

If $a \equiv b \pmod{m}$ holds **then** $c \cdot a \equiv c \cdot b \pmod{m}$, where c is any integer, holds by Theorem 5 with $d = c$.

- Adding an integer to both sides of a valid congruence preserves validity.

If $a \equiv b \pmod{m}$ holds then $c + a \equiv c + b \pmod{m}$, where c is any integer, holds by Theorem 5 with $d = c$.

- Dividing a congruence by an integer does not always produce a valid congruence.

Example: The congruence $14 \equiv 8 \pmod{6}$ holds. But dividing both sides by 2 does not produce a valid congruence since $14/2 = 7$ and $8/2 = 4$, but $7 \not\equiv 4 \pmod{6}$.

See Section 4.3 for conditions when division is ok.

Computing the mod m Function of Products and Sums

- We use the following corollary to Theorem 5 to compute the remainder of the product or sum of two integers when divided by m from the remainders when each is divided by m .

Corollary: Let m be a positive integer and let a and b be integers. Then

$$(a + b) \text{ (mod } m) = ((a \text{ mod } m) + (b \text{ mod } m)) \text{ mod } m$$

and

$$ab \text{ mod } m = ((a \text{ mod } m) (b \text{ mod } m)) \text{ mod } m.$$

(proof in text)

ARITHMETIC MODULO m

Definitions: Let Z_m be the set of nonnegative integers less than m : $\{0, 1, \dots, m-1\}$

- The operation $+_m$ is defined as $a +_m b = (a + b) \bmod m$. This is *addition modulo m* .
- The operation $\cdot_m$ is defined as $a \cdot_m b = (a \cdot b) \bmod m$. This is *multiplication modulo m* .
- Using these operations is said to be doing *arithmetic modulo m* .

Example: Find $7 +_{11} 9$ and $7 \cdot_{11} 9$.

Solution: Using the definitions above:

- $7 +_{11} 9 = (7 + 9) \bmod 11 = 16 \bmod 11 = 5$
- $7 \cdot_{11} 9 = (7 \cdot 9) \bmod 11 = 63 \bmod 11 = 8$

ARITHMETIC MODULO M

- The operations $+_m$ and $\cdot_m$ satisfy many of the same properties as ordinary addition and multiplication.
 - *Closure*: If a and b belong to $\mathbf{Z}_m$, then $a +_m b$ and $a \cdot_m b$ belong to $\mathbf{Z}_m$.
 - *Associativity*: If a , b , and c belong to $\mathbf{Z}_m$, then $(a +_m b) +_m c = a +_m (b +_m c)$ and $(a \cdot_m b) \cdot_m c = a \cdot_m (b \cdot_m c)$.
 - *Commutativity*: If a and b belong to $\mathbf{Z}_m$, then $a +_m b = b +_m a$ and $a \cdot_m b = b \cdot_m a$.
 - *Identity elements*: The elements 0 and 1 are identity elements for addition and multiplication modulo m , respectively.
 - If a belongs to $\mathbf{Z}_m$, then $a +_m 0 = a$ and $a \cdot_m 1 = a$. continued $\rightarrow$

ARITHMETIC MODULO M

- *Additive inverses:* If $a \neq 0$ belongs to $\mathbf{Z}_m$, then $m - a$ is the additive inverse of a modulo m and 0 is its own additive inverse.
 - $a +_m (m - a) = 0$ and $0 +_m 0 = 0$
- *Distributivity:* If a , b , and c belong to $\mathbf{Z}_m$, then
 - $a \cdot_m (b +_m c) = (a \cdot_m b) +_m (a \cdot_m c)$ and $(a +_m b) \cdot_m c = (a \cdot_m c) +_m (b \cdot_m c)$. (a
- Exercises 42-44 ask for proofs of these properties.
- Multiplicative inverses have not been included since they do not always exist. For example, there is no multiplicative inverse of 2 modulo 6 .
- (*optional*) Using the terminology of abstract algebra, $\mathbf{Z}_m$ with $+_m$ is a commutative group and $\mathbf{Z}_m$ with $+_m$ and $\cdot_m$ is a commutative ring.

- Does 17 divide each of these numbers?
a) 68 b) 84 c) 357 d) 1001
- Prove that if a is an integer other than 0, then
a) 1 divides a . b) a divides 0.
- Prove that part (ii) of Theorem 1 is true.
- Prove that part (iii) of Theorem 1 is true.
- Show that if $a \mid b$ and $b \mid a$, where a and b are integers, then $a = b$ or $a = -b$.
- Show that if a, b, c , and d are integers, where $a \neq 0$, such that $a \mid c$ and $b \mid d$, then $ab \mid cd$.
- Show that if a, b , and c are integers, where $a \neq 0$ and $c \neq 0$, such that $ac \mid bc$, then $a \mid b$.
- Prove or disprove that if $a \mid bc$, where a, b , and c are positive integers and $a \neq 0$, then $a \mid b$ or $a \mid c$.
- What are the quotient and remainder when
a) 19 is divided by 7?
b) -111 is divided by 11?
c) 789 is divided by 23?
d) 1001 is divided by 13?
e) 0 is divided by 19?
f) 3 is divided by 5?
g) -1 is divided by 3?
h) 4 is divided by 1?
- What are the quotient and remainder when
a) 44 is divided by 8?
b) 777 is divided by 21?
c) -123 is divided by 19?
d) -1 is divided by 23?
e) -2002 is divided by 87?
f) 0 is divided by 17?
g) 1,234,567 is divided by 1001?
h) -100 is divided by 101?
- What time does a 12-hour clock read
a) 80 hours after it reads 11:00?
b) 40 hours before it reads 12:00?
c) 100 hours after it reads 6:00?
- What time does a 24-hour clock read
a) 100 hours after it reads 2:00?
b) 45 hours before it reads 12:00?
c) 168 hours after it reads 19:00?
- Suppose that a and b are integers, $a \equiv 4 \pmod{13}$, and $b \equiv 9 \pmod{13}$. Find the integer c with $0 \leq c \leq 12$ such that
a) $c \equiv 9a \pmod{13}$.
b) $c \equiv 11b \pmod{13}$.
c) $c \equiv a + b \pmod{13}$.
d) $c \equiv 2a + 3b \pmod{13}$.
e) $c \equiv a^2 + b^2 \pmod{13}$.
f) $c \equiv a^3 - b^3 \pmod{13}$.
- Suppose that a and b are integers, $a \equiv 11 \pmod{19}$, and $b \equiv 3 \pmod{19}$. Find the integer c with $0 \leq c \leq 18$ such that
a) $c \equiv 13a \pmod{19}$.
b) $c \equiv 8b \pmod{19}$.
c) $c \equiv a - b \pmod{19}$.
d) $c \equiv 7a + 3b \pmod{19}$.
e) $c \equiv 2a^2 + 3b^2 \pmod{19}$.
f) $c \equiv a^3 + 4b^3 \pmod{19}$.
- Let m be a positive integer. Show that $a \equiv b \pmod{m}$ if $a \bmod m = b \bmod m$.
- Let m be a positive integer. Show that $a \bmod m = b \bmod m$ if $a \equiv b \pmod{m}$.
- Show that if n and k are positive integers, then $\lceil n/k \rceil = \lfloor (n-1)/k \rfloor + 1$.
- Show that if a is an integer and d is an integer greater than 1, then the quotient and remainder obtained when a is divided by d are $\lfloor a/d \rfloor$ and $a - d\lfloor a/d \rfloor$, respectively.
- Find a formula for the integer with smallest absolute value that is congruent to an integer a modulo m , where m is a positive integer.
- Evaluate these quantities.
a) $-17 \bmod 2$ b) $144 \bmod 7$
c) $-101 \bmod 13$ d) $199 \bmod 19$
- Evaluate these quantities.
a) $13 \bmod 3$ b) $-97 \bmod 11$
c) $155 \bmod 19$ d) $-221 \bmod 23$
- Find $a \operatorname{div} m$ and $a \bmod m$ when
a) $a = -111, m = 99$.
b) $a = -9999, m = 101$.
c) $a = 10299, m = 999$.
d) $a = 123456, m = 1001$.

Integer Representations and Algorithms

Section 4.2

Section Summary

- Integer Representations
 - Base b Expansions
 - Binary Expansions
 - Octal Expansions
 - Hexadecimal Expansions
- Base Conversion Algorithm
- Algorithms for Integer Operations

REPRESENTATIONS OF INTEGERS

- In the modern world, we use *decimal*, or *base 10*, notation to represent integers. For example when we write 965, we mean $9 \cdot 10^2 + 6 \cdot 10^1 + 5 \cdot 10^0$.
- We can represent numbers using any base b , where b is a positive integer greater than 1.
- The bases $b = 2$ (*binary*), $b = 8$ (*octal*), and $b = 16$ (*hexadecimal*) are important for computing and communications
- The ancient Mayans used base 20 and the ancient Babylonians used base 60.

BASE b REPRESENTATIONS

- We can use positive integer b greater than 1 as a base, because of this theorem:

Theorem 1: Let b be a positive integer greater than 1. Then if n is a positive integer, it can be expressed uniquely in the form:

$$n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b + a_0$$

where k is a nonnegative integer, $a_0, a_1, \dots, a_k$ are nonnegative integers less than b , and $a_k \neq 0$. The a_j , $j = 0, \dots, k$ are called the base- b digits of the representation.

(We will prove this using mathematical induction in Section 5.1.)

- The representation of n given in Theorem 1 is called the *base b expansion* of n and is denoted **by** $(a_k a_{k-1} \dots a_1 a_0)_b$.
- We usually omit the subscript 10 for base 10 expansions.

BINARY EXPANSIONS

Most computers represent integers and do arithmetic with binary (base 2) expansions of integers. In these expansions, the only digits used are 0 and 1.

Example: What is the decimal expansion of the integer that has $(1\ 0101\ 1111)_2$ as its binary expansion?

Solution:

$$(1\ 0101\ 1111)_2 = 1 \cdot 2^8 + 0 \cdot 2^7 + 1 \cdot 2^6 + 0 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 351.$$

Example: What is the decimal expansion of the integer that has $(11011)_2$ as its binary expansion?

Solution: $(11011)_2 = 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 27.$

OCTAL EXPANSIONS

The octal expansion (base 8) uses the digits $\{0,1,2,3,4,5,6,7\}$.

Example: What is the decimal expansion of the number with octal expansion $(7016)_8$?

Solution: $7 \cdot 8^3 + 0 \cdot 8^2 + 1 \cdot 8^1 + 6 \cdot 8^0 = 3598$

Example: What is the decimal expansion of the number with octal expansion $(111)_8$?

Solution: $1 \cdot 8^2 + 1 \cdot 8^1 + 1 \cdot 8^0 = 64 + 8 + 1 = 73$

HEXADECIMAL EXPANSIONS

The hexadecimal expansion needs 16 digits, but our decimal system provides only 10. So letters are used for the additional symbols. The hexadecimal system uses the digits $\{0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F\}$. The letters A through F represent the decimal numbers 10 through 15.

Example: What is the decimal expansion of the number with hexadecimal expansion $(2AE0B)_{16}$?

Solution:

$$2 \cdot 16^4 + 10 \cdot 16^3 + 14 \cdot 16^2 + 0 \cdot 16^1 + 11 \cdot 16^0 = 175627$$

Example: What is the decimal expansion of the number with hexadecimal expansion $(E5)_{16}$?

Solution: $14 \cdot 16^1 + 5 \cdot 16^0 = 224 + 5 = 229$

BASE CONVERSION

Example: Find the octal expansion of $(12345)_{10}$

Solution: Successively dividing by 8 gives:

- $12345 = 8 \cdot 1543 + 1$
- $1543 = 8 \cdot 192 + 7$
- $192 = 8 \cdot 24 + 0$
- $24 = 8 \cdot 3 + 0$
- $3 = 8 \cdot 0 + 3$

The remainders are the digits from right to left yielding $(30071)_8$.

Exercises

- Convert the decimal expansion of each of these integers to a binary expansion.
a) 231 b) 4532 c) 97644
- Convert the decimal expansion of each of these integers to a binary expansion.
a) 321 b) 1023 c) 100632
- Convert the binary expansion of each of these integers to a decimal expansion.
a) $(1\ 1111)_2$ b) $(10\ 0000\ 0001)_2$
c) $(1\ 0101\ 0101)_2$ d) $(110\ 1001\ 0001\ 0000)_2$
- Convert the binary expansion of each of these integers to a decimal expansion.
a) $(1\ 1011)_2$ b) $(10\ 1011\ 0101)_2$
c) $(11\ 1011\ 1110)_2$ d) $(111\ 1100\ 0001\ 1111)_2$
- Convert the octal expansion of each of these integers to a binary expansion.
a) $(572)_8$ b) $(1604)_8$
c) $(423)_8$ d) $(2417)_8$
- Convert the binary expansion of each of these integers to an octal expansion.
a) $(1111\ 0111)_2$
b) $(1010\ 1010\ 1010)_2$
c) $(111\ 0111\ 0111\ 0111)_2$
d) $(101\ 0101\ 0101\ 0101)_2$
- Convert the hexadecimal expansion of each of these integers to a binary expansion.
a) $(80E)_{16}$ b) $(135AB)_{16}$
c) $(ABBA)_{16}$ d) $(DEFACED)_{16}$
- essary, and translating each block of three binary digits into a single octal digit.
- Show that the binary expansion of a positive integer can be obtained from its octal expansion by translating each octal digit into a block of three binary digits.
- Convert $(7345321)_8$ to its binary expansion and $(10\ 1011\ 1011)_2$ to its octal expansion.
- Give a procedure for converting from the hexadecimal expansion of an integer to its octal expansion using binary notation as an intermediate step.
- Give a procedure for converting from the octal expansion of an integer to its hexadecimal expansion using binary notation as an intermediate step.
- Explain how to convert from binary to base 64 expansions and from base 64 expansions to binary expansions and from octal to base 64 expansions and from base 64 expansions to octal expansions.
- Find the sum and the product of each of these pairs of numbers. Express your answers as a binary expansion.
a) $(100\ 0111)_2, (111\ 0111)_2$
b) $(1110\ 1111)_2, (1011\ 1101)_2$
c) $(10\ 1010\ 1010)_2, (1\ 1111\ 0000)_2$
d) $(10\ 0000\ 0001)_2, (11\ 1111\ 1111)_2$
- Find the sum and product of each of these pairs of numbers. Express your answers as a base 3 expansion.
a) $(112)_3, (210)_3$
b) $(2112)_3, (12021)_3$
c) $(20001)_3, (1111)_3$

Primes and Greatest Common Divisors

Section 4.3

SECTION SUMMARY

- Prime Numbers and their Properties
- Conjectures and Open Problems About Primes
- Greatest Common Divisors and Least Common Multiples
- The Euclidian Algorithm
- gcds as Linear Combinations

PRIMES

Definition: A positive integer p greater than 1 is called *prime* if the only positive factors of p are 1 and p . A positive integer that is greater than 1 and is not prime is called *composite*.

Example: The integer 7 is prime because its only positive factors are 1 and 7, but 9 is composite because it is divisible by 3.

The Fundamental Theorem of Arithmetic

Theorem: Every positive integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of no decreasing size.

Examples:

- [illegible]

GREATEST COMMON DIVISOR

Definition: Let a and b be integers, not both zero. The largest integer d such that $d \mid a$ and also $d \mid b$ is called the greatest common divisor of a and b . The greatest common divisor of a and b is denoted by $\gcd(a, b)$.

One can find greatest common divisors of small numbers by inspection.

Example: What is the greatest common divisor of 24 and 36?

Solution: $\gcd(24, 36) = 12$

Example: What is the greatest common divisor of 17 and 22?

Solution: $\gcd(17, 22) = 1$

GREATEST COMMON DIVISOR

Definition: The integers a and b are *relatively prime* if their greatest common divisor is 1.

Example: 17 and 22

Definition: The integers $a_1, a_2, \dots, a_n$ are *pairwise relatively prime* if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.

Example: Determine whether the integers 10, 17 and 21 are pairwise relatively prime.

Solution: Because $\gcd(10, 17) = 1$, $\gcd(10, 21) = 1$, and $\gcd(17, 21) = 1$, 10, 17, and 21 are pairwise relatively prime.

Example: Determine whether the integers 10, 19, and 24 are pairwise relatively prime.

Solution: Because $\gcd(10, 24) = 2$, 10, 19, and 24 are not pairwise relatively prime.

GREATEST COMMON DIVISOR

Definition: The integers a and b are *relatively prime* if their greatest common divisor is 1.

Example: 17 and 22

Definition: The integers $a_1, a_2, \dots, a_n$ are *pairwise relatively prime* if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.

Example: Determine whether the integers 10, 17 and 21 are pairwise relatively prime.

Solution: Because $\gcd(10, 17) = 1$, $\gcd(10, 21) = 1$, and $\gcd(17, 21) = 1$, 10, 17, and 21 are pairwise relatively prime.

Example: Determine whether the integers 10, 19, and 24 are pairwise relatively prime.

Solution: Because $\gcd(10, 24) = 2$, 10, 19, and 24 are not pairwise relatively prime.

Finding the Greatest Common Divisor Using Prime Factorizations

- Suppose the prime factorizations of a and b are:

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}, \quad b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n},$$

where each exponent is a nonnegative integer, and where all primes occurring in either prime factorization are included in both. Then:

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_n^{\min(a_n, b_n)}.$$

- This formula is valid since the integer on the right (of the equals sign) divides both a and b . No larger integer can divide both a and b .

Example: $120 = 2^3 \cdot 3 \cdot 5$ $500 = 2^2 \cdot 5^3$

$$\gcd(120, 500) = 2^{\min(3, 2)} \cdot 3^{\min(1, 0)} \cdot 5^{\min(1, 3)} = 2^2 \cdot 3^0 \cdot 5^1 = 20$$

- Finding the gcd of two positive integers using their prime factorizations is not efficient because there is no efficient algorithm for finding the prime factorization of a positive integer.

LEAST COMMON MULTIPLE

Definition: The least common multiple of the positive integers a and b is the smallest positive integer that is divisible by both a and b . It is denoted by $\text{lcm}(a,b)$.

- The least common multiple can also be computed from the prime factorizations.

$$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \cdots p_n^{\max(a_n, b_n)}$$

This number is divided by both a and b and no smaller number is divided by a and b .

Example: $\text{lcm}(2^3 3^5 7^2, 2^4 3^3) = 2^{\max(3,4)} 3^{\max(5,3)} 7^{\max(2,0)} = 2^4 3^5 7^2$

- The greatest common divisor and the least common multiple of two integers are related by:

Theorem 5: Let a and b be positive integers. Then

$$ab = \text{gcd}(a,b) \cdot \text{lcm}(a,b)$$

EUCLIDEAN ALGORITHM

- The Euclidian algorithm is an efficient method for computing the greatest common divisor of two integers. It is based on the idea that $\gcd(a,b)$ is equal to $\gcd(a,c)$ when $a > b$ and c is the remainder when a is divided by b .

Example: Find $\gcd(91, 287)$:

- $287 = 91 \cdot 3 + 14$

Divide 287 by 91

- $91 = 14 \cdot 6 + 7$

Divide 91 by 14

- $14 = 7 \cdot 2 + 0$

Divide 14 by 7

Stopping
condition

$$\gcd(287, 91) = \gcd(91, 14) = \gcd(14, 7) = 7$$

continued →

FINDING GCDS AS LINEAR COMBINATIONS

Example: Express $\gcd(252, 198) = 18$ as a linear combination of 252 and 198.

Solution: First use the Euclidean algorithm to show $\gcd(252, 198) = 18$

i. $252 = 1 \cdot 198 + 54$

ii. $198 = 3 \cdot 54 + 36$

iii. $54 = 1 \cdot 36 + 18$

iv. $36 = 2 \cdot 18$

- Now working backwards, from iii and i above
 - $18 = 54 - 1 \cdot 36$
 - $36 = 198 - 3 \cdot 54$
 - Substituting the 2nd equation into the 1st yields:
 - $18 = 54 - 1 \cdot (198 - 3 \cdot 54) = 4 \cdot 54 - 1 \cdot 198$
 - Substituting $54 = 252 - 1 \cdot 198$ (from i) yields:
 - $18 = 4 \cdot (252 - 1 \cdot 198) - 1 \cdot 198 = 4 \cdot 252 - 5 \cdot 198$
- This method illustrated above is a two pass method. It first uses the Euclidean algorithm to find the gcd and then works backwards to express the gcd as a linear combination of the original two integers. A one pass method, called the *extended Euclidean algorithm*, is developed in the exercises.

Exercises

1. Determine whether each of these integers is prime.
 - a) 21 b) 29
 - c) 71 d) 97
 - e) 111 f) 143
 2. Determine whether each of these integers is prime.
 - a) 19 b) 27
 - c) 93 d) 101
 - e) 107 f) 113
 3. Find the prime factorization of each of these integers.
 - a) 88 b) 126 c) 729
 - d) 1001 e) 1111 f) 909,090
 4. Find the prime factorization of each of these integers.
 - a) 39 b) 81 c) 101
 - d) 143 e) 289 f) 899
 5. Find the prime factorization of $10!$.
-
32. Use the Euclidean algorithm to find
 - a) $\gcd(1, 5)$. b) $\gcd(100, 101)$.
 - c) $\gcd(123, 277)$. d) $\gcd(1529, 14039)$.
 - e) $\gcd(1529, 14038)$. f) $\gcd(11111, 111111)$.
 33. Use the Euclidean algorithm to find
 - a) $\gcd(12, 18)$. b) $\gcd(111, 201)$.
 - c) $\gcd(1001, 1331)$. d) $\gcd(12345, 54321)$.
 - e) $\gcd(1000, 5040)$. f) $\gcd(9888, 6060)$.
-
25. What are the greatest common divisors of these pairs of integers?
 - a) $3^7 \cdot 5^3 \cdot 7^3, 2^{11} \cdot 3^5 \cdot 5^9$
 - b) $11 \cdot 13 \cdot 17, 2^9 \cdot 3^7 \cdot 5^5 \cdot 7^3$
 - c) $23^{31}, 23^{17}$
 - d) $41 \cdot 43 \cdot 53, 41 \cdot 43 \cdot 53$
 - e) $3^{13} \cdot 5^{17}, 2^{12} \cdot 7^{21}$
 - f) 1111, 0
 26. What is the least common multiple of each pair in Exercise 24?
 27. What is the least common multiple of each pair in Exercise 25?
 28. Find $\gcd(1000, 625)$ and $\text{lcm}(1000, 625)$ and verify that $\gcd(1000, 625) \cdot \text{lcm}(1000, 625) = 1000 \cdot 625$.
 29. Find $\gcd(92928, 123552)$ and $\text{lcm}(92928, 123552)$, and verify that $\gcd(92928, 123552) \cdot \text{lcm}(92928, 123552) = 92928 \cdot 123552$. [Hint: First find the prime factorizations of 92928 and 123552.]
 30. If the product of two integers is $2^7 3^8 5^2 7^{11}$ and their greatest common divisor is $2^3 3^4 5$, what is their least common multiple?

SOLVING CONGRUENCES

Section 4.4

SECTION SUMMARY

- Linear Congruences
- The Chinese Remainder Theorem
- Computer Arithmetic with Large Integers (*not currently included in slides, see text*)
- Fermat's Little Theorem
- Pseudoprimes
- Primitive Roots and Discrete Logarithms

LINEAR CONGRUENCES

Definition: A congruence of the form

$$ax \equiv b \pmod{m},$$

where m is a positive integer, a and b are integers, and x is a variable, is called a *linear congruence*.

- The solutions to a linear congruence $ax \equiv b \pmod{m}$ are all integers x that satisfy the congruence.

Definition: An integer $\bar{a}$ such that $\bar{a}a \equiv 1 \pmod{m}$ is said to be an *inverse* of a modulo m .

Example: 5 is an inverse of 3 modulo 7 since $5 \cdot 3 = 15 \equiv 1 \pmod{7}$

- One method of solving linear congruencies makes use of an inverse $\bar{a}$, if it exists. Although we can not divide both sides of the congruence by a , we can multiply by $\bar{a}$ to solve for x .

INVERSE OF a MODULO m

- The following theorem guarantees that an inverse of a modulo m exists whenever a and m are relatively prime. Two integers a and b are relatively prime when $\gcd(a,b) = 1$.

Theorem 1: If a and m are relatively prime integers and $m > 1$, then an inverse of a modulo m exists. Furthermore, this inverse is unique modulo m . (This means that there is a unique positive integer $\bar{a}$ less than m that is an inverse of a modulo m and every other inverse of a modulo m is congruent to $\bar{a}$ modulo m .)

Proof: Since $\gcd(a,m) = 1$, by Theorem 6 of Section 4.3, there are integers s and t such that $sa + tm = 1$.

- Hence, $sa + tm \equiv 1 \pmod{m}$.
- Since $tm \equiv 0 \pmod{m}$, it follows that $sa \equiv 1 \pmod{m}$.
- Consequently, s is an inverse of a modulo m .
- The uniqueness of the inverse is Exercise 7.



FINDING INVERSES

- The Euclidean algorithm and Bézout coefficients gives us a systematic approaches to finding inverses.

Example: Find an inverse of 3 modulo 7.

Solution: Because $\gcd(3,7) = 1$, by Theorem 1, an inverse of 3 modulo 7 exists.

- Using the Euclidian algorithm: $7 = 2 \cdot 3 + 1$.
- From this equation, we get $-2 \cdot 3 + 1 \cdot 7 = 1$, and see that -2 and 1 are Bézout coefficients of 3 and 7.
- Hence, -2 is an inverse of 3 modulo 7.
- Also every integer congruent to -2 modulo 7 is an inverse of 3 modulo 7, i.e., 5, -9 , 12, etc.

Finding Inverses

Example: Find an inverse of 101 modulo 4620.

Solution: First use the Euclidian algorithm to show that $\gcd(101, 4620) = 1$.

Working Backwards:

$$42620 = 45 \cdot 101 + 75$$

$$101 = 1 \cdot 75 + 26$$

$$75 = 2 \cdot 26 + 23$$

$$26 = 1 \cdot 23 + 3$$

$$23 = 7 \cdot 3 + 2$$

$$3 = 1 \cdot 2 + 1$$

$$2 = 2 \cdot 1$$

$$1 = 3 - 1 \cdot 2$$

$$1 = 3 - 1 \cdot (23 - 7 \cdot 3) = -1 \cdot 23 + 8 \cdot 3$$

$$1 = -1 \cdot 23 + 8 \cdot (26 - 1 \cdot 23) = 8 \cdot 26 - 9 \cdot 23$$

$$1 = 8 \cdot 26 - 9 \cdot (75 - 2 \cdot 26) = 26 \cdot 26 - 9 \cdot 75$$

$$1 = 26 \cdot (101 - 1 \cdot 75) - 9 \cdot 75$$

$$= 26 \cdot 101 - 35 \cdot 75$$

$$1 = 26 \cdot 101 - 35 \cdot (42620 - 45 \cdot 101)$$

$$= -35 \cdot 42620 + 1601 \cdot 101$$

Since the last nonzero remainder is 1,
 $\gcd(101, 4260) = 1$

Bézout coefficients : -35 and 1601

1601 is an inverse of
101 modulo 42620

Using Inverses to Solve Congruences

- We can solve the congruence $ax \equiv b \pmod{m}$ by multiplying both sides by $\bar{a}$.

Example: What are the solutions of the congruence $3x \equiv 4 \pmod{7}$.

Solution: We found that -2 is an inverse of 3 modulo 7 (two slides back). We multiply both sides of the congruence by -2 giving

$$-2 \cdot 3x \equiv -2 \cdot 4 \pmod{7}.$$

Because $-6 \equiv 1 \pmod{7}$ and $-8 \equiv 6 \pmod{7}$, it follows that if x is a solution, then $x \equiv -8 \equiv 6 \pmod{7}$

We need to determine if every x with $x \equiv 6 \pmod{7}$ is a solution. Assume that $x \equiv 6 \pmod{7}$. By Theorem 5 of Section 4.1, it follows that $3x \equiv 3 \cdot 6 = 18 \equiv 4 \pmod{7}$ which shows that all such x satisfy the congruence.

The solutions are the integers x such that $x \equiv 6 \pmod{7}$, namely, $6, 13, 20 \dots$ and $-1, -8, -15, \dots$

Exercises

1. Show that 15 is an inverse of 7 modulo 26.
-  2. Show that 937 is an inverse of 13 modulo 2436.
3. By inspection (as discussed prior to Example 1), find an inverse of 4 modulo 9.
4. By inspection (as discussed prior to Example 1), find an inverse of 2 modulo 17.
5. Find an inverse of a modulo m for each of these pairs of relatively prime integers using the method followed in Example 2.
 - a) $a = 4, m = 9$
 - b) $a = 19, m = 141$
 - c) $a = 55, m = 89$
 - d) $a = 89, m = 232$
6. Find an inverse of a modulo m for each of these pairs of relatively prime integers using the method followed in Example 2.
 - a) $a = 2, m = 17$
 - b) $a = 34, m = 89$
 - c) $a = 144, m = 233$
 - d) $a = 200, m = 1001$
- *7. Show that if a and m are relatively prime positive integers, then the inverse of a modulo m is unique modulo m . [*Hint:* Assume that there are two solutions b and c of the congruence $ax \equiv 1 \pmod{m}$. Use Theorem 7 of Section 4.3 to show that $b \equiv c \pmod{m}$.]
8. Show that an inverse of a modulo m , where a is an integer and $m > 2$ is a positive integer, does not exist if $\gcd(a, m) > 1$.
9. Solve the congruence $4x \equiv 5 \pmod{9}$ using the inverse of 4 modulo 9 found in part (a) of Exercise 5.
10. Solve the congruence $2x \equiv 7 \pmod{17}$ using the inverse of 2 modulo 7 found in part (a) of Exercise 6.
11. Solve each of these congruences using the modular inverses found in parts (b), (c), and (d) of Exercise 5.
 - a) $19x \equiv 4 \pmod{141}$
 - b) $55x \equiv 34 \pmod{89}$
 - c) $89x \equiv 2 \pmod{232}$